

Values variables and types

Project: Python readiness portfolio

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

From a paper algorithm to Python

Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace a saved value for Values variables and types. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Values variables and types. Proposed explanation: “Digits always mean a number”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Trace values, types and reassignment

Predict both output lines and explain why estimate stays unchanged when count is reassigned. Repeat the trace with initial count=0. Explain why note is text even though its printed characters look like a number.

```
count = 4
note = "4"
estimate = count / 2
fits = count <= 5
count = count + 3
print(count, note, estimate, fits)
print(type(count).__name__, type(note).__name__, type(estimate).__name__,
type(fits).__name__)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Identify the value and its type

Classify 12, 1.5, "12", False and "False". Explain why the last two are different.

My reasoning and evidence

6. W2 Trace a saved value

Trace the five-line count and saved program. Record both printed outputs, then predict the result of assigning saved = count after the change. Check with a run or label a paper trace.

My reasoning and evidence

7. W3 Diagnose the operation

Predict $6 + 3$, $"6" + "3"$ and $"6" * 3$. Explain what happens for $"6" + 3$. Which expression counts six groups of three?

My reasoning and evidence

8. W4 Trace a new pair of names independently

A fresh script runs `seats = 12; booked = seats; seats = seats - 5; print(seats); print(booked)`, with each statement on a separate line. Predict both outputs. Then add `booked = seats + 1` and `print(booked)`. Explain the evaluation order and choose a type for a seat label `"07"`.

My reasoning and evidence

9. W5 Separate type from validity

Give a suitable type for a counter count, a length of 2.5 metres, a shelf ID with a leading zero, and a yes-or-no availability flag. Give a count value with the right type that should still be rejected.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled values variables and types evidence sheet to Python readiness portfolio. Show how you can distinguish integer, floating-point, text and boolean values. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week’s record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record:

Evidence: First step I repaired and why

My record:

Evidence: New case checked later; date and result

My record:

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Values variables and types

A number and its printed-looking text. A program stores a numeric badge count and a string that looks like that count.

Python 3: `count=8; tag="8"; print(count+2); print(tag+"2")`.

Values variables and types

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Inspect types

count is an integer; tag is a string.



2. Apply addition

Integer addition gives 10; string concatenation gives the text 82.



3. Predict a mixed operation

count+tag raises TypeError rather than silently converting tag.

CHANGE ONE THING

Change count to 10, leaving tag unchanged.

Explain the changed result using the matching lesson.

Predict both print outputs and explain why they differ. Does count+tag work, and what happens to each valid output after count becomes ten?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

13. Project portfolio milestone

Project: Python readiness portfolio

Current milestone: Python readiness portfolio. Keep runnable code, predictions, actual outputs, one repaired error and an explanation of a function on changed inputs.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A number and its printed-looking text. A program stores a numeric badge count and a string that looks like that count.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? `count+tag` raises `TypeError` rather than silently converting `tag`.

1. Predict a mixed operation
2. Inspect types
3. Apply addition

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: `count+tag` raises `TypeError` rather than silently converting `tag`. Values carry types, not merely appearances. The plus operator performs numeric addition for these integers and concatenation for these strings; the mixed operation is invalid here.

Practice 2: explain the deciding evidence

Changed-case practice: Change `count` to 10, leaving `tag` unchanged. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. The two valid outputs become 12 and 82.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The two valid outputs become 12 and 82. Only the integer variable changes; the string still contains the character 8.

Repair a missing building block

Review From a paper algorithm to Python (python-bridge/1). Goal: Translate a short paper calculation into a Python script. Short practice: Match each paper quantity to its Python name. Why does the spare count get added once, and what prints when spare is five?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
